VACUUM RESERVOIR

DESCRIPTION

The vacuum reservoir is a plastic storage tank connected to an engine vacuum source by vacuum lines.

OPERATION

The vacuum reservoir is used to supply the vacuum needed to maintain proper speed control operation when engine vacuum drops, such as in climbing a grade while driving. A one-way check valve is used in the vacuum line between the reservoir and the vacuum source. This check valve is used to trap engine vacuum in the reservoir. On certain vehicle applications, this reservoir is shared with the heating/air-conditioning system. The vacuum reservoir cannot be repaired and must be replaced if faulty.

DIAGNOSIS AND TESTING - VACUUM RESERVOIR

(1) Disconnect vacuum hose at speed control servo and install a vacuum gauge into the disconnected hose.

(2) Start engine and observe gauge at idle. Vacuum gauge should read at least ten inches of mercury.

(3) If vacuum is less than ten inches of mercury, determine source of leak. Check vacuum line to engine for leaks. Also check actual engine intake manifold vacuum. If manifold vacuum does not meet this requirement, check for poor engine performance and repair as necessary.

(4) If vacuum line to engine is not leaking, check for leak at vacuum reservoir. To locate and gain access to reservoir, refer to Vacuum Reservoir Removal/Installation in this group. Disconnect vacuum line at reservoir and connect a hand-operated vacuum pump to reservoir fitting. Apply vacuum. Reservoir vacuum should not bleed off. If vacuum is being lost, replace reservoir.

(5) Verify operation of one-way check valve and check it for leaks.

(a) Locate one-way check valve. The valve is located in vacuum line between vacuum reservoir and engine vacuum source. Disconnect vacuum hoses (lines) at each end of valve.

(b) Connect a hand-operated vacuum pump to reservoir end of check valve. Apply vacuum. Vacuum should not bleed off. If vacuum is being lost, replace one-way check valve.

(c) Connect a hand-operated vacuum pump to vacuum source end of check valve. Apply vacuum. Vacuum should flow through valve. If vacuum is not flowing, replace one-way check valve. Seal the fitting at opposite end of valve with a finger and apply vacuum. If vacuum will not hold, diaphragm within check valve has ruptured. Replace valve.

REMOVAL

The vacuum reservoir is located under the vehicle battery tray (Fig. 6).

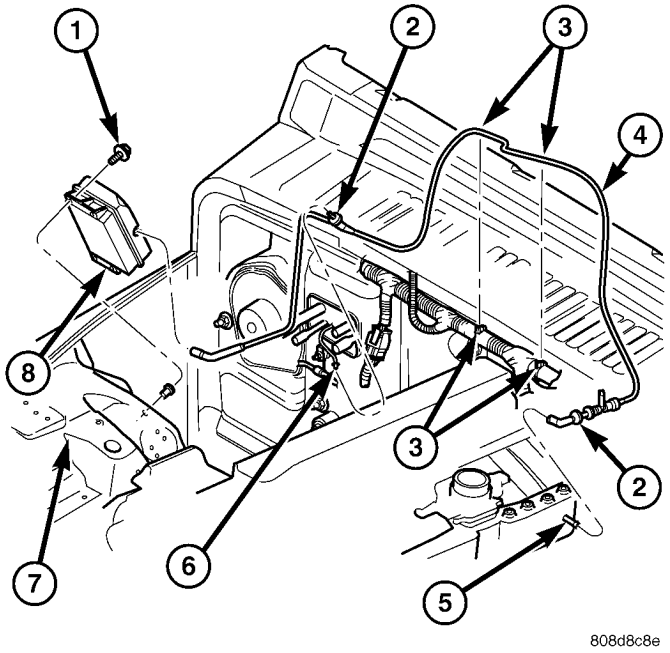

Fig. 6 Vacuum Reservoir Removal/Installation

1 - SCREW
2 - VACUUM CHECK VALVE (2)
3 - VACUUM LINE RETAINER (2)
4 - ACCESSORY VACUUM SUPPLY LINE
5 - ENGINE VACUUM FITTING
6 - HVAC VACUUM SUPPLY LINE
7 - INNER FENDER
8 - VACUUM RESERVOIR

(1) Remove battery and battery tray. Refer to Battery Removal/Installation.

(2) Disconnect vacuum supply line at reservoir (Fig. 6).

(3) Remove screw securing reservoir to inner fender.

(4) Remove reservoir from vehicle.

INSTALLATION

The vacuum reservoir is located under the vehicle battery tray (Fig. 6).

(1) Position reservoir to vehicle and install mounting screw.

(2) Tighten screw to 1.2 N·m (10 in. lbs.) torque.

(3) Connect vacuum line to reservoir.

(4) Install battery and battery tray. Refer to Battery Removal/Installation.